COMSATS University Islamabad
Department of Computer Science
Course Description Form (CDF)

Course Information

Course Code: **CSC336**	Course Title: **Web Technologies**
Credit Hours: **3(2,1)**	Lecture Hours/Week: **2**
Lab Hours/Week: **3**	Pre-Requisites: **CSC241–Object Oriented Programming**

Course Objectives

- To explain the fundamental concepts of web architectures and its applications;
- To provide an understanding of planning, designing and publishing a multi-page website;
- To provide hands-on experience of client-side technologies such as HTML, JavaScript, CSS, and React.js
- To provide practical experience in developing dynamic web applications using modern full-stack technologies like Fast API, PostgreSQL, React.

Course Contents

This course introduces the modern web technologies used for web development. Topics include Overview of Web Platforms; Web Architectures; Markup Languages; Styling; Client-Side Languages; Server-Side Technologies; Use of Databases in Web-based Applications; REST APIs; and Programming under Platform Constraints.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	Web Development: Overview, Platforms, Terminologies, Standards & Constraints, Applications Categories, and Development Environments. Model-View-Controller Architecture, Tiered Web Architecture: One-Tier, Two-Tier, and N-Tier.	1
2.	Markup Languages: HTML, HTML5, Block Level vs. Inline Elements, semantic HTML, HTML Forms	2
3.	Styling: Cascading Style Sheets (CSS), CSS3, Layout Designs, Responsive and Adaptive Layouts, Bootstrap/Tailwind Framework	6
4.	Modern JavaScript (ES6+) for Web Development: Core JavaScript concepts (variables, data types, operators, scope), Functions, Closures, and Higher-Order Functions, Important built-in methods (map, filter, reduce, forEach), DOM Manipulation & Event Handling, Asynchronous JavaScript (callbacks, promises, async/await), Regular Expressions for validation and parsing	6
5.	Front-end development using React.js: Introduction to React.js; JSX in React, Rendering Approaches), React Components, Event Handling, Hooks, State Management, Routing.	7
6.	Full-Stack Development using FastAPI & PostgreSQL: Introduction to Python-based Web Development & FastAPI, REST APIs, JSON, Fetch/Axios API, FastAPI for backend API development, PostgreSQL as relational database (CRUD operations, schemas, migrations), SQLAlchemy / Pydantic for ORM and data validation.	8

Total Contact Hours	**30**

Mapping of CLOs and Gas

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO's for Theory				
CLO-1	1-2	Classify web architectures and related applications.	*Understanding*	2
CLO-2	3-5	Demonstrate the capabilities for developing professional front-end using client-side technologies.	*Applying*	3-5
CLO-3	6	Illustrate the concepts of server-side technologies for secure database interactions.	*Applying*	3-5
CLO's for Lab				
CLO-4	3-6	Apply the concepts of markup & scripting languages and client-side technologies.	*Applying*	3-5
CLO-5	2-6	Develop dynamic applications using current industrial practices.	*Creating*	1, 3-6

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	-	Quiz 1 & 2	Quiz 3 & 4	-	-
Assignments	-	Assignment 1&2	Assignment 3&4	Lab Assignments	Lab Assignments
Midterm Exam	Mid Term Exam	Mid Term Exam	-	Mid Term Exam	-
Final Term Exam	Final Term Exam			-	Lab Project/ Lab Final Term Exam

Text and Reference Books

Textbooks:

1. Web Design Playground: HTML & CSS the Interactive Way, Paul McFedries, Manning, 2019.
2. Eloquent JavaScript by Marijn Haverbeke (4th edition, 2024)
3. Learning React: Modern Patterns for Developing React Apps 2nd Edition by Alex Banks, Eve Porcello, 2020.
4. Learn React Hooks by Daniel Bugl, 2019.
5. FastAPI (O'Reilly).

Reference Books:

1. Web Engineering, Sahil Rai, Kuk University Notes, 2020.
2. Web Programming with HTML5, CSS, and JavaScript, John Dean, Jones & Bartlett Learning 2018.
3. Murach's Modern JavaScript by Mary Delamater (2024)
4. Mastering FastAPI: Build High-Performance APIs with Python — Aimed at building scalable, high-performance APIs with advanced features.